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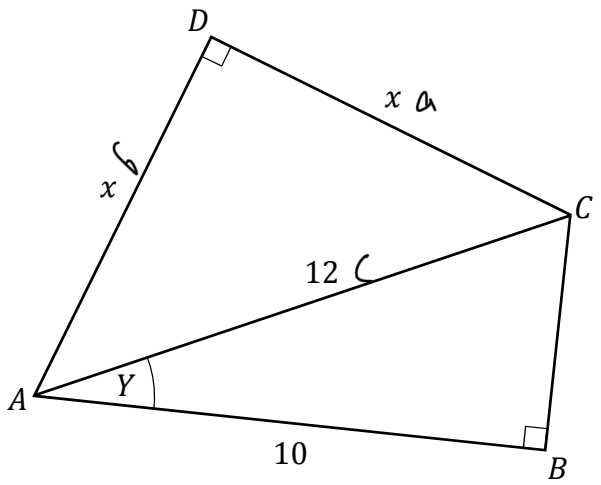
Question 8 **(Suggested maximum time: 15 minutes)**

(a) The diagram below shows two right-angled triangles, ABC and ACD.
They have right angles at B and D, respectively.
$|AB| = 10$, $|AC| = 12$, and $|AD| = |DC| = x$, as shown.
The angle BAC is marked Y.



(i) Use trigonometry to find the size of the angle Y.
Give your answer correct to one decimal place.

$\cos Y = 10/12$

$\cos^{-1}(10/12) = Y$

$Y = 33.6°$

(ii) Find the value of x. Give your answer correct to two decimal places.

$a^2 + b^2 = 12^2$

$x^2 + x^2 = 144$

$2x^2 = 144$

$x^2 = 72$

$x = 8.49$ cm

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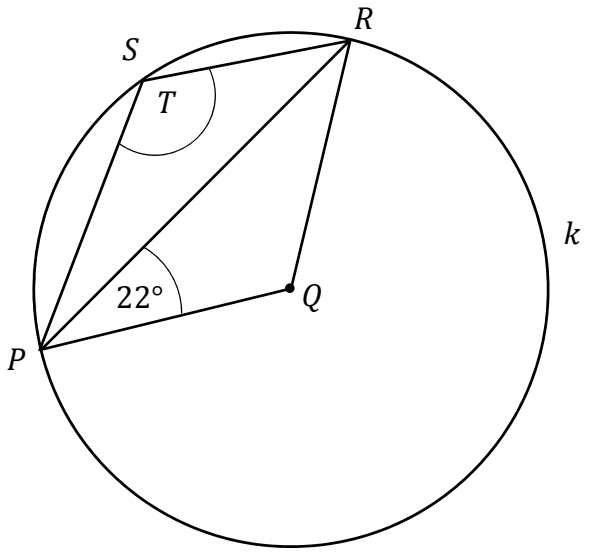
(b) The diagram below shows two different triangles, PQR and RSP.
The points R, S, and P are on the circle k, and Q is the centre of the circle k.
$|\angle QPR| = 22°$, as shown. The angle PSR is marked T.



Find the size of the angle T. Show all of your working out.

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